SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – NUMERICAL HACKATHON

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	**Assessment:**	Assessment Tool 3 – HACKATHON
Weightage:	30 % of CIA	**Total Marks:**	100
CO3 Statement:	*Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)*		
Student Name:	Pavithra . R	**Reg. No.:**	192421369
Section / Batch:	Slot - D	**Date:**	11/08/2026

Code of Conduct

I Pavithra.R (192421369) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Pavithra R

Instructions

- Duration: 60 minutes
- Number of questions : 10

Faculty In-charge	**Course Coordinator**	**Head of Department**
Terrance Frederick	Terrance Frederick	~~Terrance Frederick~~
Signature:	Signature:	Signature:
Date: 11/08/26	Date: 11/08/26	Date:

85/100

CO3 - AT3 Hackathon.

Pavithra. R
192421369
11/8/2026.

1. step 1: Identify the given information.
 * Tunneling Over: 20 bytes per packet.
 * Data packet size: 500 bytes.

step 2: Understand the Goal
You need to find the additional overhead introduced by tunnel for this specific 500-byte packet.

solution:- 500 bytes of data + 20 bytes of overhead.

= 520 bytes.

1.A) The additional overhead for the 500 byte data packet is 20 bytes.

2. step 1: convert Bandwidth:

$20MHz = 20 \times 10^6 Hz$.

step 2: Find the Symbol
symbol rate = 20×10^6 symbol/sec.

step 3: calculate data rate.
Each symbol carries 6 bits.
Data rate = $20 \times 10^6 \times 6$
$= 120 \times 10^6$ bits/sec

120 Mbps.

QAM - Amplitude & phase.

3. step 1:

No. of entries : 50

size of each entry : 32 byte

step 2: calculate the total Memo

50 entries x 32 bytes/entry

= 1600 bytes.

the total Memory Required is

1600 bytes or 1.6 kilo bytes.

4. step: 1

speed = 10 m/s

update Interval = 2 seconds.

Step: 2

Formula:

Distance = Speed x Time

= 10 x 2

= 20 Meters.

5. step: 1

DSR, a Route Request packet is forward from one node to the next until it Reaches the Destination.

Source → Intermediate 1 → 2 → 3 → 4 → 5 → Destinati

5 Intermediate + 1 source

= 6.

subnet mask: 255.255.255.240

The last octet 240 in binary is

240 = 11110000.

So the subnet Mask is :

/28.

Step 1: Find Host bits.

IPV4 has 32 bits total.

32 - 28 = 4 host bits.

Step 2: Find usable IP addresses

2^4 = 16 total addresses.

But 2 Addresses are Reserved.

=> 1 network Address

=> 2 Broadcast Address

16 - 2 = 14.

∴ 14 mobile nodes can be assigned IP address in one /28 IPV4 subnet.

=> IPV4 => 4 parts called octets => 8 bits.

=> subnet mask => which part of the IP is for the Network. and which part is for devices

7. Step 1:

Channel bandwidth = 20MHz

Symbol Rate = 1MSps = 1 million symbols/second

Step 2:

Formula:

For spectral efficiency = Date Rate / Bandwidth

Need bits per symbol to find the data rate.

Data Rate = 1MSps × 1 = 1 Mbps

∴ spectral efficiency = 1Mbps / 20MHz

= 0.05 bits/s/Hz.

$$h = \frac{\text{Symbol rate} \times \text{Bits/symbol}}{\text{Bandwidth}}$$

If they give a modulation like 64 QAM, the bits/symbol = 6,

8. Step 1:- No. of nodes = 50

Network area = 1km2

transmission Range = 250m

$$\text{Network Density} = \frac{\text{No. of nodes}}{\text{Area}}$$

Area = $\pi \times r^2$

$\pi \times 250^2$

$$= \frac{50}{1}$$

= 50 nodes / km2

Step 1:

velocity = 30 m/s

Frequency (f) = 900 MHz (900×10^6 Hz)

Speed of light (c) = a standard constant of 3×10^8 m/s.

$$f_d = \frac{v \times f}{c}.$$

$$f_d = \frac{30 \times (900 \times 10^6)}{3 \times 10^8}$$

$$f_d = \frac{27000 \times 10^6}{3 \times 10^8}$$

$$f_d = \frac{27 \times 10^9}{3 \times 10^8} = 90 \text{ Hz}$$

10.

Step 1:

Data Rate: 54 Mbps

packet size: 1500 bytes.